



Lecture 5: HVDC Transmission

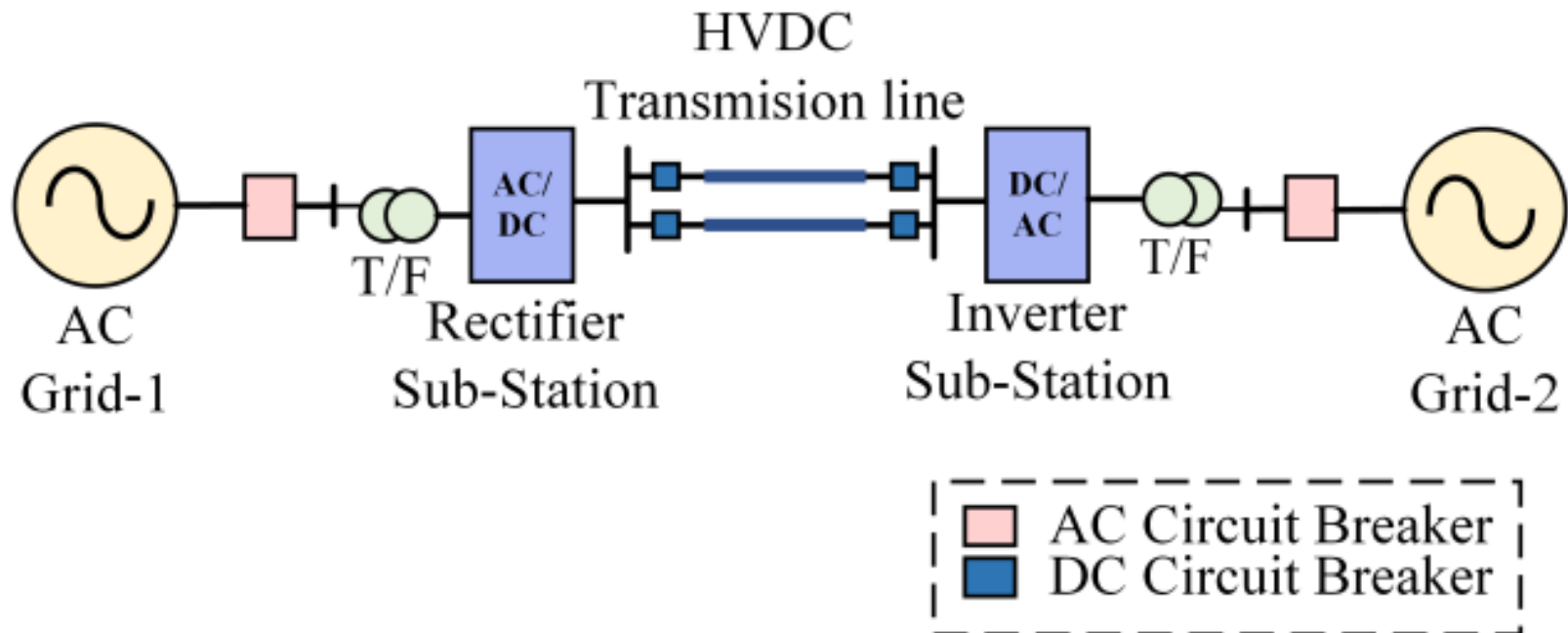
EEE 321: Power System I

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Presentation Outline

- ❑ Introduction to HVDC transmission
- ❑ Advantages and Disadvantages of HVDC transmission
- ❑ Types of HVDC links
- ❑ Converter technologies used in HVDC
- ❑ Examples

Introduction to HVDC transmission



The first commercial HVDC scheme was connected between the island of Gotland to the Sweden mainland in 1954. The system was built within a 96 km distance at 100 KV potential and a capacity of 20 MW.

During the subsequent 55 years, great advances in HVDC technology and economic opportunities for HVDC have been achieved.

Advantages and Disadvantages of HVDC

Advantages of HVDC Transmission:

☐ **Reduced Transmission Losses:**

HVDC lines have lower losses than HVAC lines over long distances due to the absence of reactive power loss and skin effect.

☐ **Ability to Transmit Power Undersea:**

HVDC is particularly well-suited for submarine cable applications where AC transmission is limited by capacitance and charging currents.

☐ **Interconnection of Asynchronous AC Systems:**

HVDC can connect grids operating at different frequencies or with different phase angles, which is crucial for integrating renewable energy sources and connecting different power systems.

☐ **Increased Capacity of Existing Lines:**

HVDC can transmit more power through the same right-of-way as an equivalent HVAC system, increasing the capacity of existing transmission infrastructure.

☐ **Enhanced Stability and Control:**

HVDC systems offer better control over power flow, enabling faster response to grid disturbances and improved stability, especially in large-scale renewable energy integration.

☐ **Lower Electromagnetic Interference:**

HVDC produces less electromagnetic interference than HVAC, making it suitable for sensitive environments.

Advantages and Disadvantages of HVDC

Disadvantages of HVDC Transmission:

❑ Higher Initial Costs:

HVDC systems require expensive conversion equipment (rectifiers and inverters) at both ends of the transmission line, increasing initial investment costs.

❑ More Complex Maintenance:

HVDC equipment, including converters and control systems, is more complex than HVAC equipment, requiring specialized maintenance and expertise.

❑ Lower Reliability in Multi-Terminal Configurations:

While HVDC is reliable for point-to-point transmission, multi-terminal HVDC systems can be more complex and prone to reliability issues.

❑ Limited Availability and Capacity:

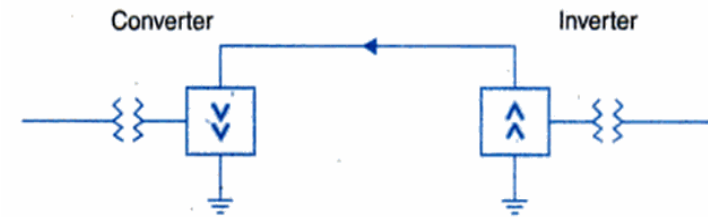
HVDC technology is not as widely deployed as HVAC, and some applications may have capacity limitations compared to HVAC systems.

Types of HVDC Links

Monopolar HVDC Link

A monopolar HVDC link is the simplest and most economical configuration.

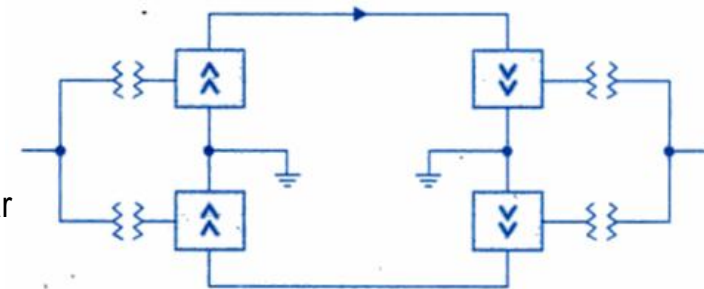
- ❑ Single high-voltage conductor, typically with a negative polarity.
- ❑ The return path for the current is through the ground, sea, or a dedicated metallic conductor.
- ❑ The entire system fails if a fault occurs on the single conductor, as there is no redundancy.



Bipolar HVDC Link

Bipolar links are the most common type of HVDC system for long-distance, high-power transmission.

- ❑ It uses two conductors, one with a positive polarity and the other with a negative polarity, both operating at the same voltage with respect to ground.
- ❑ Under normal operating conditions, the currents in the two poles are balanced and flow in opposite directions, resulting in zero ground current.
- ❑ If a fault occurs on one pole, the system can automatically switch to a monopolar mode using the ground as a return path, allowing the remaining pole to continue supplying half the rated power.



There is another type called Homopolar HVDC link, which is rarely used today due to their reliance on ground return currents.

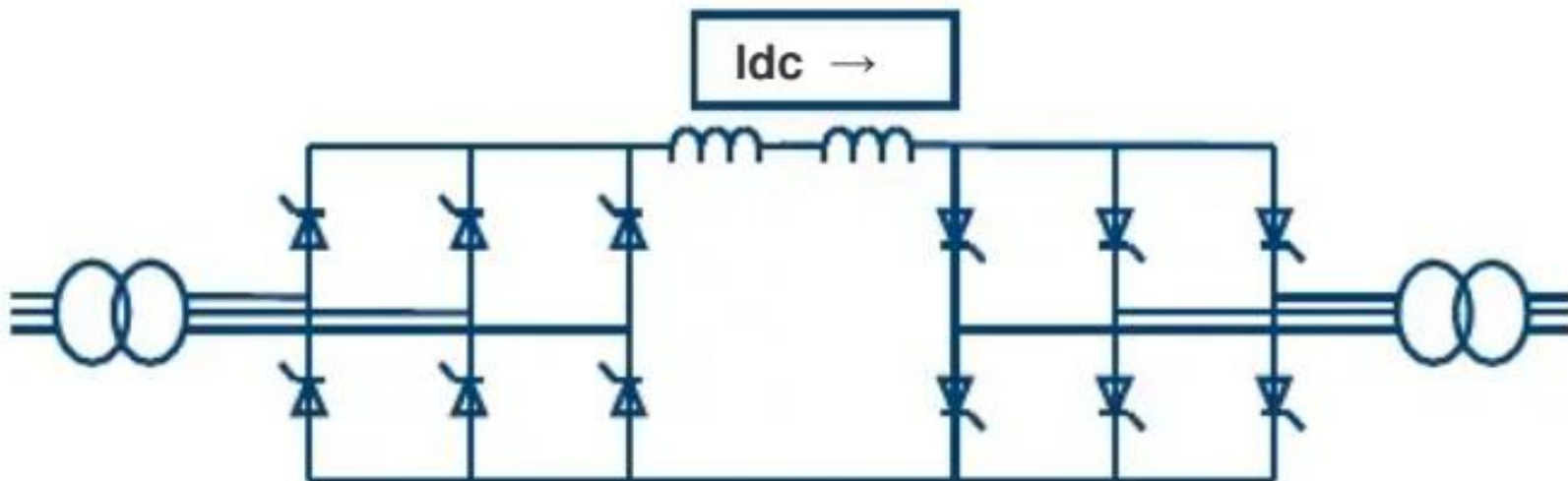
Converter Technologies used in HVDC

HVDC power transmission systems can be classified into 2 categories based on converter technologies that are given below:

1. Line commutated converter (LCC) based HVDC transmission system
2. Voltage source converter (VSC) based HVDC transmission system

LCC Based HVDC Transmission

LCC-HVDC is developed based on thyristors. A thyristor is a current-controlled device (hence the other name is Current source converter CSC-HVDC) and In LCC-HVDC transmission system, the active power flow is controlled by adjusting the firing angle (turn-on). The typical smoothing inductance value on the DC side for a large HVDC system is in the range of 0.1 H to 0.5 H. In LCC-HVDC, the converter requires reactive power of around 50 - 60 % of the converter power rating and these are supplied by the filters and capacitor banks connected on the AC side.



Converter Technologies used in HVDC

Advantages of LCC-HVDC Transmission Systems

- ❑ **High voltage and power rating** (for example, power levels up to 6400 MW).
- ❑ Low losses because thyristor has **low on-state losses** and negligible switching losses. The losses in one terminal are about 0.7 % of the rated current.
- ❑ LCC-HVDC has a low rate of rise of DC fault current because of large inductance on the DC side.

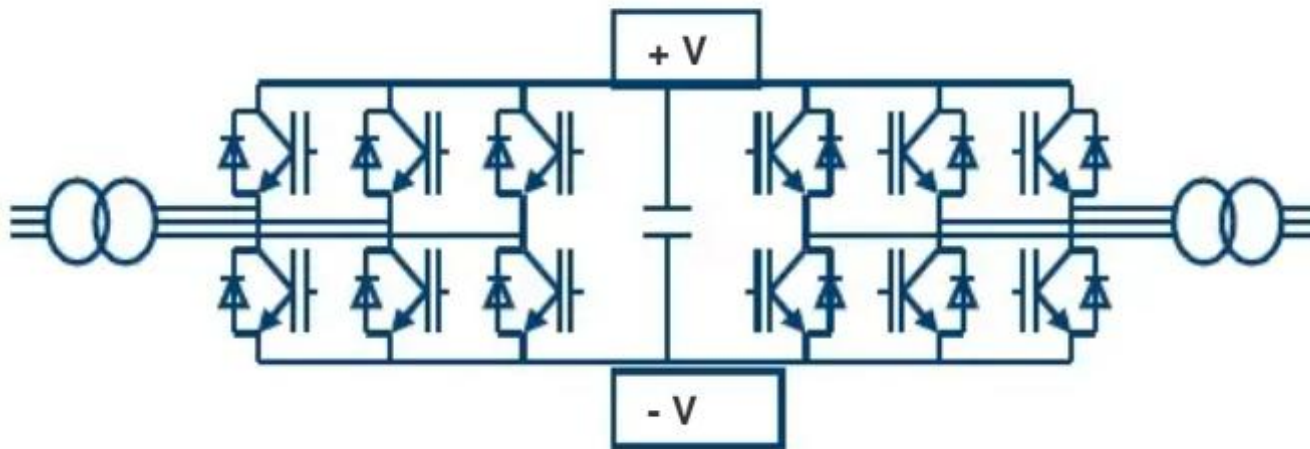
Limitations of LCC-HVDC Transmission Systems

- ❑ The power flow is unidirectional. The reversal of active power flow requires a change in the DC link voltage polarity of the system. So, building multi-terminal HVDC grid is difficult.
- ❑ **It cannot be operated with weak AC systems (Low short circuit ratio $SCR < 2$), making it unsuitable for connecting to weak or remote AC networks.**
- ❑ **Independent control of active and reactive power is not possible.**

Converter Technologies used in HVDC

VSC based HVDC transmission

The VSC-based HVDC transmission system is developed based on self-commutating devices such as Insulated Gate Bipolar Transistor (IGBT). The IGBT is a voltage-controlled device and therefore it is called a voltage source converter (VSC) based HVDC system. The self-commutating devices allow the application of high-frequency (over 1 kHz) pulse-width modulation (PWM) techniques that allow the precise control of active and reactive powers by controlling the full AC system voltage.



Converter Technologies used in HVDC

Advantages of VSC-HVDC Transmission Systems

- ❑ VSC-HVDC has active and reactive power control independently. So, the VSC-HVDC can be operated with weak AC systems (short circuit ratio $SCR < 2$). VSC can generate leading and lagging reactive power independently and so it can provide voltage support while transmitting active power at any level.
- ❑ If there is no power transmission in VSC-HVDC, both converter stations can be operated independently as STATCOM to provide reactive power/voltage support to the local AC networks.
- ❑ With the use of PWM (with a switching frequency in kHz), the harmonic filters are at higher frequencies. Hence, it reduces the size, losses, and cost.
- ❑ Black-start capability: VSC can start or restore the power to the AC network without power generation units.

Limitations of VSC-HVDC Transmission Systems

- ❑ VSC-HVDC is vulnerable to DC side faults. The rate of rise of the DC fault current is higher because of the low reactor on the DC side. The 2-level VSC behaves like an uncontrollable diode bridge during DC side faults.
- ❑ It gives higher losses than LCC-HVDC because the IGBTs in VSC-HVDC have high on-state losses (i.e., the losses in one VSC terminal are about 1.6 %). High-frequency switching in VSC-HVDC can lead to high switching losses.
- ❑ Lower power ratings because IGBTs in VSC-HVDC have lower power capability than thyristors in LCC-HVDC. IGBTs in VSC-HVDC have lower current overload capability than thyristors in LCC-HVDC.

Converter Technologies used in HVDC

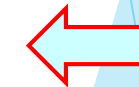
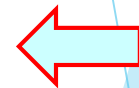
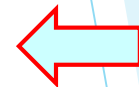
Comparison between LCC-HVDC and VSC-HVDC Transmission Systems

LCC HVDC

- Use semiconductors which can withstand voltage in either polarity
- Output voltage can be either polarity to change power direction
- Current direction does not change
- Store energy inductively
- Use semiconductors which can turn on by control action
- Turn-off and “commutation” rely on the external circuit
- Lower station losses
- Lower cost
- Higher reliability
- More mature technology

VSC HVDC

- Use semiconductors which can pass current in either direction
- Output voltage polarity does not change
- Current direction changes to change Power direction
- Store energy capacitively
- Use semiconductors which can turn on or off by control action
- Turn-off is independent of external circuit
- Higher station losses
- Higher cost by 10 – 15%
- Lower reliability, due to high component count
- Less mature technology

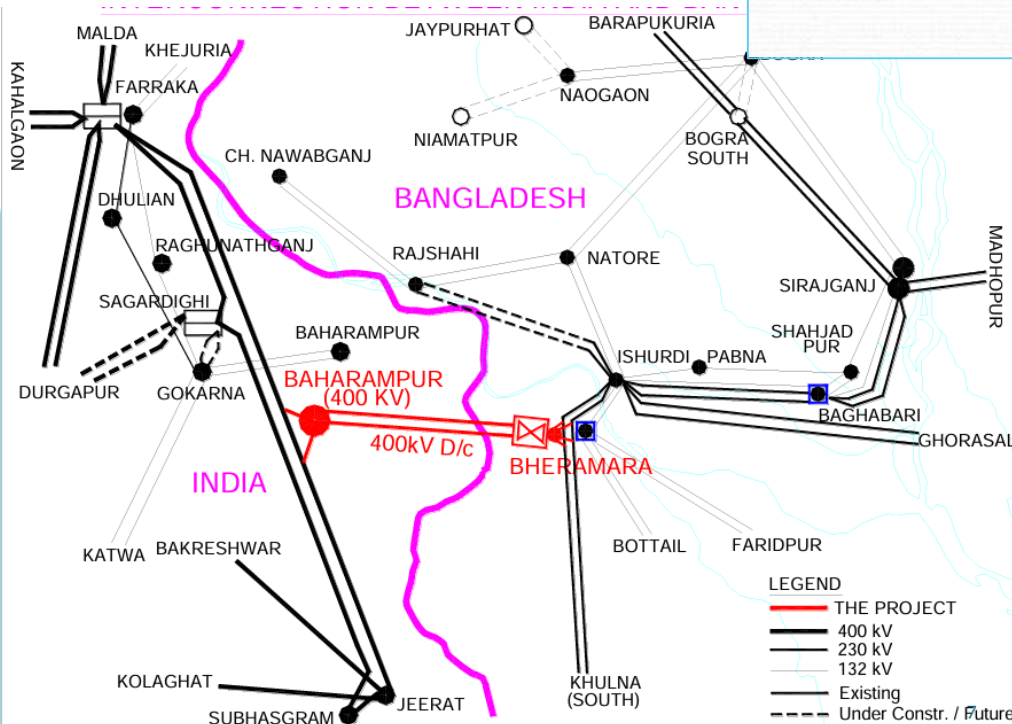
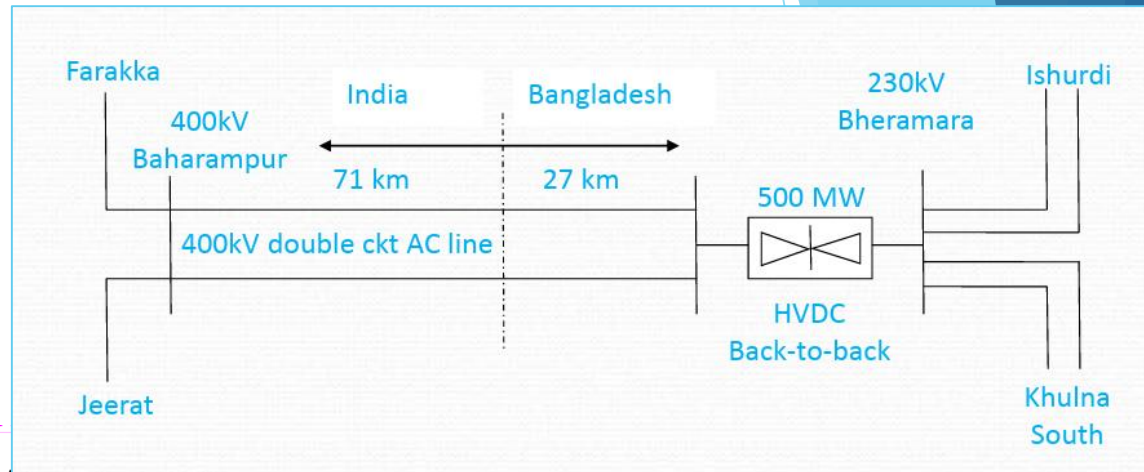


Example of HVDC System

This project, the first of its kind in South Asia, connects the eastern Indian grid to the western Bangladeshi grid via a HVDC back-to-back link. The Bheramara substation has two converter stations (Block 1 and Block 2), with a combined capacity of 1000 MW. From Bheramara at the Western border of Bangladesh, the 230-kilovolt (kV) grid of the country is connected via a substation and overhead lines to India's 400-kV grid.



Example of HVDC System



Reference: Electrical Power Systems, C.L.Wadhwa, 4th edition, New Age International.



QUESTIONS AND DISCUSSIONS

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